IBM HR Analysis Project

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
print("Import Success")
```

```
Import Success
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
df = pd.read_csv('/content/drive/My Drive/IBM HR Data Analysis/WA_Fn-UseC_-HR-Employee-Attrition.csv
```

```
print(df.head())
```

```
df.shape
```

```
df.info()
```

```
df.describe()
```

Pre Processing.

1.Remove unwanted columns . because these columns have single value or consistent or does not needed in analysis. columns are . Employee count , over 18 , Employee Number , standard hours

```
col_to_drop = ['EmployeeCount','Over18','EmployeeNumber','StandardHours']
df = df.drop(col_to_drop,axis=1)
```

```
df.isnull().sum()
```

```
df.duplicated().sum()
```

```
df['Department'] = df['Department'].replace({"Research & Development":"R & D" , "Human Resources":
```

```
df['Expected_Dailyrate'] = df['HourlyRate'] * 8
```

EDA

```
relation = df.select_dtypes(include="number").corr()
strong_relation = relation [(relation < -0.2)|(relation > 0.2)]
strong_relation
```

```
! pip install ydata_profiling
```

```
from ydata_profiling import ProfileReport
```

```
profile = ProfileReport(df ,title = "EDA of IBM HR Data Analysis " , explorative=True)
profile
```

EDA

Question 1 :What is the overall attrition rate, and how does it compare across departments?

```
# Total attrition Counts
attrition_rate = df['Attrition'].value_counts()
sns.barplot(x=attrition_rate.index,y=attrition_rate.values , palette="viridis")
```

```
#Department wise Propotion .
attrition_propotion = df.groupby('Department')['Attrition'].value_counts(normalize=True)
attrition_propotion.plot(kind = "bar" , stacked = True)
plt.xlabel("Department")
plt.ylabel("Attrition Propotion")
plt.title("Department wise Attrition Propotion")
# sns.histplot(data=df, x="Department", hue="Attrition", multiple="fill" , palette="viridis")
# plt.xlabel("Department")
# plt.ylabel("Proportion")
```

Question 2 : Which job roles have the highest and lowest attrition rates?

```
Attrition_jobrole = df.groupby('JobRole')["Attrition"].value_counts().unstack().sort_values(by="Yes
print(Attrition_jobrole)
```

so accroding to our stats Laboratry Technician has highest Job attrition rate and Research Director has less job attrition rate.

```
Attrition_jobrole.plot(kind="bar" , stacked=True)
```

Question 3 : Do employees who work overtime leave at a significantly higher rate than those who don't?

```
Attrition_overtime = df.groupby('OverTime')["Attrition"].value_counts(normalize=True).unstack().sor
Attrition_overtime.plot(kind="bar" , stacked=True)
```

```
Attrition_overtime
```

if we look at the above results 30% of employess left who work ovetime and 10 % employees left who donot do overtime . people left the company because ovetime may disturb their worklife balance.

Question 4 :How does business travel frequency affect the likelihood of attrition?

```
BusinessTravel_attrition = df.groupby('BusinessTravel')["Attrition"].value_counts(normalize=True).u
BusinessTravel_attrition =BusinessTravel_attrition * 100
BusinessTravel_attrition
```

```
BusinessTravel_attrition.plot(kind="bar" , stacked=True)
```

if we look at above stats the employees who travel frequently have high attrition rate upto 24 % ann on the other hand who travel rarely have nearly 15 % but non-Travel have only 8% rate.

Question 5 : Is marital status a significant predictor of attrition — do single employees leave more?

```
MartialStatus_attrition = df.groupby('MaritalStatus')["Attrition"].value_counts(normalize=True).uns
MartialStatus_attrition =MartialStatus_attrition * 100
MartialStatus_attrition
```

```
MartialStatus_attrition.plot(kind="bar" , stacked=True)
```

if we Look at the stats Single employees left the company have 25.5% high rate . Married employees has less rate which is 12.5% and Divorced has only 10% rate . Yes Single Employees leave more.

Question 6: What percentage of employees who left were at Job Level 1 vs senior levels

```
joblevel_attrition  = df.groupby("JobLevel")["Attrition"].value_counts(normalize=True).unstack()
joblevel_attrition = joblevel_attrition * 100
joblevel_attrition
#
```

```
joblevel_attrition.plot(kind="bar" , stacked=True)
```

If we look at the above stats employees at level 1 has attrition rate 26% while senior who are at level 5 attition rate only 7%

Question 7:At what age bracket does attrition peak — are younger employees more likely to leave?

```
age_groups = pd.cut(df['Age'] , bins=[18,30,45,60] , labels = ["Young","Middle-aged","Senior"])
age_attrition = df.groupby(age_groups)["Attrition"].value_counts(normalize=True).unstack()
age_attrition = age_attrition * 100
age_attrition
```

```
age_attrition.plot(kind="bar" , stacked=True)
```

Yes Younger employess Has high rate of attrition of 25% and while middle aged and senior employess has 12% rate of attrition.

Question 8: Does gender influence attrition rate in this organization?

```
gender_attrition = df.groupby('Gender')["Attrition"].value_counts(normalize=True).unstack()
gender_attrition = gender_attrition * 100
gender_attrition
```

Yes Males has slighlty High Rate attrition which is 17% in male and 14.7% in female .

```
gender_attrition.plot(kind="bar" , stacked=True)
```

Question 9 :Which combination of factors (OverTime + Low Income + Low Satisfaction) creates the highest attrition risk?

```
multifactor_attrition = df.groupby(['OverTime','JobSatisfaction','PerformanceRating'])['Attrition']
multifactor_attrition = multifactor_attrition * 100
multifactor_attrition
```

```
multifactor_attrition.plot(kind="bar" , stacked=True)
plt.xlabel("Overtime + JobSatisfaction + PerformanceRating ")
plt.ylabel("Attrition Rate")
plt.title("Multifactor Attrition")
```

employees with overtime , jobsatisfaction at level 1 and performance rating 4 has high attrition rate which is 50% . And employees with overtime , job satisfaction level 2 and performance rating 4 has 2nd high attrition rate which is 46%.

Question 10 : What is the attrition rate among employees who have worked at 4+ previous companies (job hoppers)?

```
workedexp_attrition = df.groupby('NumCompaniesWorked')["Attrition"].value_counts(normalize=True).ur
workedexp_attrition
```

```
workedexp_attrition.plot(kind="bar" , stacked=True)
```

The attrition rate is higher in employees who have prviously worked at more then 4 companies which is 20% +.

Question 11 Do employees with long commutes (DistanceFromHome > 20) leave more frequently?

```
df20above = df[df['DistanceFromHome']>20]
DistanceFromHome_attrition = df20above.groupby('DistanceFromHome')["Attrition"].value_counts(normal
DistanceFromHome_attrition = DistanceFromHome_attrition * 100
DistanceFromHome_attrition
```

Employees living farther from the workplace tend to show higher attrition in certain distance ranges, suggesting that long commuting distances may contribute to employee turnover.

Question 12 : How does stock option level affect attrition — do employees with no stock options leave more?

```
stockoption_attrition = df.groupby('StockOptionLevel')["Attrition"].value_counts(normalize=True).ur
stockoption_attrition = stockoption_attrition * 100
stockoption_attrition
#
```

Yes employees with no stock option has high rate of attrition 24% but employees with High stock options also high attrition rate upto ~18%

Question 13 : What is the age distribution of the workforce — is IBM's workforce aging?

```
sns.histplot(data=df['Age'],kde = True , palette='rainbow')
```

Question 14 : What is the gender breakdown across departments and job roles?

```
department_jobrole_gender = df.groupby(['Department','JobRole'])['Gender'].value_counts(normalize=1
department_jobrole_gender = department_jobrole_gender * 100
department_jobrole_gender
```

Question 15 : How does education level vary across departments?

```
department_education = df.groupby('Department')['Education'].value_counts(normalize=True).unstack(
department_education = department_education * 100
department_education
```

Question 16 :What education fields are most common in Research & Development vs Sales?

```
department_education.plot(kind="bar" , stacked=False)
```

In Both Departments Level 3 (Bechlor's) Degree is common and master degree is at second number.

Question 17 : What is the marital status distribution, and does it differ by department?

```
matrialStatus_deprtment = df.groupby('Department')['MaritalStatus'].value_counts(normalize=True).ur
matrialStatus_deprtment = matrialStatus_deprtment * 100
matrialStatus_deprtment
```

```
matrialStatus_deprtment.plot(kind="bar" , stacked=False)
```

Question 18 : What proportion of employees have never worked at another company (NumCompaniesWorked = 0)?

```
zerocompaniess = df[df['NumCompaniesWorked']==0]
propostion = (len(zerocompaniess)/len(df)) * 100
```

```
print(propostion)
```

Question 19 : Is there a correlation between age and education level in this company?

```
relation = df['Age'].corr(df['Education'])
print(relation)
```

```
education_age = df.groupby('Education')['Age'].mean()
```

```
sns.lineplot(x = education_age.index , y = education_age.values , palette='viridis', markers=".")
```

Although age increases with education level, the relationship is not very strong, indicating that employees of different ages exist across all education levels.

Question 20: What is the average monthly income by department and job role?

```
avgMonthlyincome = df.groupby(['Department','JobRole'])['MonthlyIncome'].mean()
avgMonthlyincome
```

```
avgMonthlyincome_pivit = avgMonthlyincome.unstack()
avgMonthlyincome_pivit
```

Question 21 : How does monthly income vary by job level — is the pay progression linear?

```
monthlyincome_joblevel = df.groupby('JobLevel')['MonthlyIncome'].mean()
monthlyincome_joblevel
```

```
sns.lineplot(x = monthlyincome_joblevel.index , y = monthlyincome_joblevel.values)
```

Yes This is linear Propostion . As job level increases the Monthly income is lineraly increasing.

Quesetion 22 : Do employees who left earn significantly less than those who stayed?

```
attration_monthlyincome = df.groupby('Attrition')['MonthlyIncome'].mean()
attration_monthlyincome
```

```
sns.barplot(x = attration_monthlyincome.index , y = attration_monthlyincome.values , palette="virid
```

Yes the employees who stayed earn more then the employees who left the company .Thats mean the less monthly income leads to more attrition rate.

Question 23 :What is the average salary hike % given to high performers vs standard performers?

```
performance_salaryhike  = df.groupby('PerformanceRating')['PercentSalaryHike'].mean()
performance_salaryhike
```

```
sns.barplot(x = performance_salaryhike.index , y = performance_salaryhike.values , palette="magma")
```

There is Difference between the Salary hike of standard performers and high performers . The salary hike for statndard performer is 14 % and for high performers its ~23%.

Question 24 : How does stock option level distribute across job roles — are senior roles better compensated in equity?

```
jobrole_joblevel_stockoption = df.groupby(['JobRole','JobLevel'])['StockOptionLevel'].value_counts(
jobrole_joblevel_stockoption = jobrole_joblevel_stockoption * 100
```

```
jobrole_joblevel_stockoption
```

```
jobrole_joblevel_stockoption.plot(kind="bar" , stacked=False)
```

Most employees across job roles receive StockOptionLevel 0 or 1, while levels 2 and 3 are relatively rare, indicating that higher stock option benefits are offered to a smaller proportion of employees, typically at higher job levels.

Question 25 : Is daily rate meaningfully different from hourly rate × 8? Are there pay anomalies?

```
df["payDifference"] = df["DailyRate"] - df['Expected_Dailyrate']
df["payDifference"]
```

```
sns.scatterplot(x = df['DailyRate'] , y = df['Expected_Dailyrate'] , palette="magma")
```

If there is no difference then the scatter plot should be a strigth diagonal line but the points are scattered along the whole chart thats mean there is big difference between hourly and daily rate.

Question 26 : What is the income gap between male and female employees at the same job level?

```
paygap_genderwise = df.groupby(['Gender','JobLevel'])['MonthlyIncome'].mean()
paygap_genderwise
#
```

There is slight difference between Male and female joblevel monthly income .

Question 27 : Do employees who receive higher salary hikes stay longer at the company?

```
salrayhike_stayduration = df.groupby('PercentSalaryHike')['YearsAtCompany'].mean()
salrayhike_stayduration
```

Show hidden output

The analysis examines whether employees who receive higher salary hikes stay longer in the company. The results show no strong relationship between PercentSalaryHike and YearsAtCompany. Employees receiving higher salary hikes do not necessarily have longer tenure, suggesting that salary increases alone may not significantly influence employee retention.

Question 28 : What is the distribution of job satisfaction scores across the workforce?

```
JobSatisfaction_age = df.groupby('JobSatisfaction')['Age'].mean()
JobSatisfaction_age
```

JobSatisfaction	Age
1	36.916955
2	36.985714
3	37.022624
4	36.795207

dtype: float64

```
sns.barplot(x = JobSatisfaction_age.index , y = JobSatisfaction_age.values , palette="magma")
```

Show hidden output

There is equally Workforce Distribution accross job satisfaction.

Question 29 :Which department has the lowest average environment satisfaction?

```
enviornmentSatisfaction_department = df.groupby('Department')['EnvironmentSatisfaction'].mean().sor
enviornmentSatisfaction_department
```

Department	EnvironmentSatisfaction
Sales	2.679372
HR	2.682540
R & D	2.744017

dtype: float64

There is no big difference but the lowest Environment satisfaction is in sales department.

Question 30 :

Do employees with low job satisfaction (score = 1) have a significantly higher attrition rate?

```
jobsatisfaction_attrition = df.groupby('JobSatisfaction')["Attrition"].value_counts(normalize=True)
jobsatisfaction_attrition = jobsatisfaction_attrition * 100
jobsatisfaction_attrition
```

JobSatisfaction	Attrition No	Attrition Yes
1	77.162630	22.837370
2	83.571429	16.428571
3	83.484163	16.515837
4	88.671024	11.328976

Yes the Low Jobsatisfaction (score=1) has high rate of attrition rate as compared to other scores . As job Satisfaction Score increases the attriton rate decrease.

Question 31:How does work-life balance score vary by overtime status?

```
worklife_overtime = df.groupby('OverTime')['WorkLifeBalance'].mean()
worklife_overtime
```

OverTime	WorkLifeBalance
No	2.773245
Yes	2.730769

dtype: float64

According to stats overtime time doesnot affect the work life Balance. Both have same average wokrklife balance score.

Question 32: Is there a correlation between relationship satisfaction and years with current manager?

```
relationship_yearswithmanager = df.groupby('RelationshipSatisfaction')['YearsWithCurrManager'].mear
print(relationship_yearswithmanager)
```

```
RelationshipSatisfaction
1    4.387681
2    3.927393
3    3.915033
```

```
4    4.312500
Name: YearsWithCurrManager, dtype: float64
```

There is a slight trend, but relationship satisfaction is only weakly related to the number of years with the current manager.

Question 33: Do employees who travel frequently report lower work-life balance scores?

```
worlifebalance_businessTravel = df.groupby('BusinessTravel')['WorkLifeBalance'].mean()
worlifebalance_businessTravel
```

Show hidden output

The analysis indicates that business travel does not significantly affect employees' work-life balance

Question 34 : What combination of satisfaction scores (Job + Environment + Relationship) best predicts attrition?

```
multi_combination_Attrition = df.groupby(['JobSatisfaction','EnvironmentSatisfaction','Relationship
multi_combination_Attrition = multi_combination_Attrition * 100
multi_combination_Attrition
#
```

			Attrition	No	Yes
JobSatisfaction	EnvironmentSatisfaction	RelationshipSatisfaction			
1	1	2		16.666667	83.333333
2	1	1		44.444444	55.555556
		2		55.555556	44.444444
1	1	1		56.250000	43.750000
		3		61.538462	38.461538
...	...	...		...	...
4	4	3		93.617021	6.382979
		4		95.000000	5.000000
3	3	2		95.652174	4.347826
4	2	4		96.296296	3.703704
		2		100.000000	NaN

64 rows × 2 columns

Job satisfaction level 1 , EnvironmentSatisfaction 1 and RelationshipSatisfaction 2 has highest attrition Rate 83% .

Question 35: How does job involvement score differ between employees who left vs stayed?

```
jobinvolvement_attrition = df.groupby('JobInvolvement')["Attrition"].value_counts(normalize=True).u
jobinvolvement_attrition  = jobinvolvement_attrition * 100
jobinvolvement_attrition
```

Attrition	No	Yes
JobInvolvement		
1	66.265060	33.734940
2	81.066667	18.933333
3	85.599078	14.400922
4	90.972222	9.027778

The employees who have job envolvement score 1 has high rate of attrition which is ~34% and employees who stay has less attrition rate 9% .

Question 6 : What percentage of employees received an 'Outstanding' performance rating vs 'Excellent'?

```
performance = df['PerformanceRating'].value_counts(normalize=True)
performance = performance * 100
performance
```

PerformanceRating	proportion
3	84.62585
4	15.37415

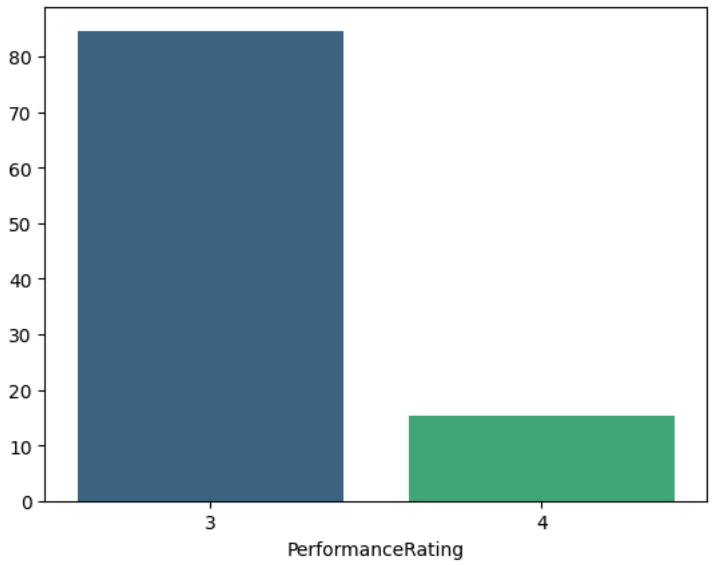
dtype: float64

```
sns.barplot(x = performance.index , y = performance.values, palette="viridis")
```

```
/tmp/ipykernel_39943/3164653402.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.barplot(x = performance.index , y = performance.values, palette="viridis")
<Axes: xlabel='PerformanceRating'>
```



Question 37: Is performance rating correlated with salary hike received?

```
performance_salaryhike = df.groupby('PerformanceRating')['PercentSalaryHike'].mean()
performance_salaryhike
```

PerformanceRating	PercentSalaryHike
3	14.003215
4	21.849558

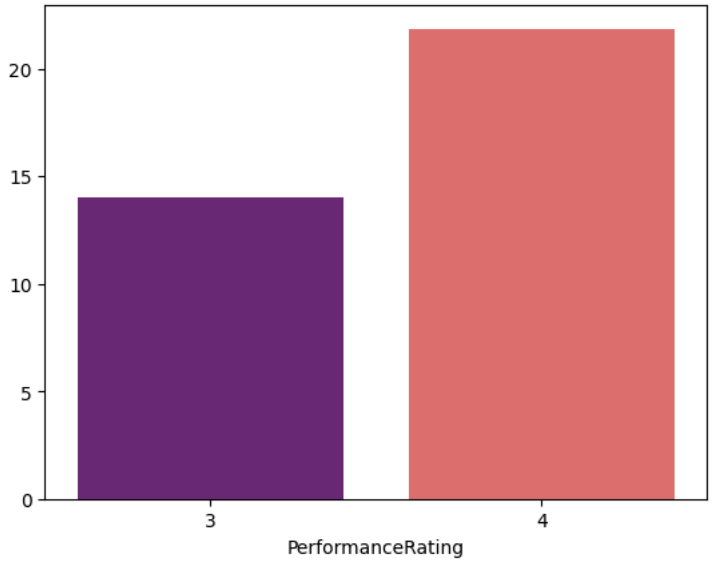
dtype: float64

```
sns.barplot(x = performance_salaryhike.index , y = performance_salaryhike.values , palette="magma")
```

```
/tmp/ipykernel_39943/1520636392.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.barplot(x = performance_salaryhike.index , y = performance_salaryhike.values , palette="magma")
<Axes: xlabel='PerformanceRating'>
```



Question 38 : How many training sessions do high performers attend vs average performers?

```
tarining_performance = df.groupby('PerformanceRating')['TrainingTimesLastYear'].mean()
tarining_performance
```

PerformanceRating	TrainingTimesLastYear
3	2.807878
4	2.752212

dtype: float64

Both have almost attend same average training session but Excellent performance rater has slightly high training last year.

Question 39 : Do high performers leave more (flight risk) or stay more than average performers?

```
performance_attrition = df.groupby('PerformanceRating')["Attrition"].value_counts(normalize=True).u
performance_attrition = performance_attrition * 100
performance_attrition
```

Attrition	No	Yes
PerformanceRating		
3	83.922830	16.077170
4	83.628319	16.371681

Double-click (or enter) to edit

```
df.info()
```

Show hidden output